

tf.compat.v1.assign_sub Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/assign_sub

Update ref by subtracting value from it.

Function Signatures

```
tf.compat.v1.assign_sub( ref, value, use_locking=None,  
name=None )
```

Code Examples

```
tf.compat.v1.assign_sub(  
    ref, value, use_locking=None, name=None  
)
```

```
tf.compat.v1.assign_sub(  
    ref, value, use_locking=None, name=None  
)
```

```
with tf.Graph().as_default():  
    with tf.compat.v1.Session() as sess:  
        a = tf.compat.v1.Variable(1, dtype=tf.int64)  
        sess.run(a.initializer)  
        update_op = tf.compat.v1.assign_sub(a, 1)  
        res_a = sess.run(update_op)  
        res_a  
0
```

```
with tf.Graph().as_default():
```

```
with tf.compat.v1.Session() as sess:
```

```
a = tf.compat.v1.Variable(1, dtype=tf.int64)
```

```
sess.run(a.initializer)
```

```
update_op = tf.compat.v1.assign_sub(a, 1)
```

Documentation

Update ref by subtracting value from it.

Migrate to TF2

`tf.compat.v1.assign_sub` is mostly compatible with eager execution and `tf.function`.

To switch to the native TF2 style, one could use method `'assign_sub'` of `tf.Variable`:

How to Map Arguments

Before & After Usage Example

Before:

After:

Description

This operation outputs ref after the update is done.

This makes it easier to chain operations that need to use the reset value.

Unlike `tf.math.subtract`, this op does not broadcast. ref and value must have the same shape.

Returns